Ignacio Quintero Mächler

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Professional Experience

Chargé de Recherche CNRS *Paris, France*
Institut de Biologie de l'École Normale Supérieure *Oct 2023 – present*

CNRS postdoctoral fellow *Paris, France*
(Dr. H. Morlon lab) Institut de Biologie de l'École Normale Supérieure *Aug 2022 – Sept 2023*

Marie Skłodowska-Curie postdoctoral fellow *Paris, France*
(Dr. H. Morlon lab) Institut de Biologie de l'École Normale Supérieure *June 2020 – Aug 2022*

CNRS postdoctoral fellow *Paris, France*
(Dr. H. Morlon lab) Institut de Biologie de l'École Normale Supérieure *Jan 2019 – May 2020*

Statistics Consultant, Yale University Library *New Haven, CT, USA*
Yale University *Feb 2017 – Oct 2018*

Education

Yale University *New Haven, CT, USA*
PhD in Ecology and Evolutionary Biology (Dr. W. Jetz lab) *Sept 2013 – Nov 2018*

Yale University *New Haven, CT, USA*
MS in Ecology and Evolutionary Biology (Dr. W. Jetz lab) *Sept 2013 – May 2016*

Universidad de los Andes *Bogotá, Colombia*
B.Sc. in Biology (Dr. C. D. Cadena lab) *Sept 2007 – June 2011*

Peer-reviewed Publications

Loss of macroevolutionary species fitness explains the rise and fall of clades 2025
Quintero, I., Andréoletti, J., Silvestro, D., Morlon, H.
*Nature Ecology & Evolution **9**, 2346–2357* 10.1038/s41559-025-02873-7

The diffused evolutionary dynamics of morphological novelty 2025
Quintero, I.
*Proceedings of the National Academy of Sciences **122** (18) e2425573122* 10.1073/pnas.2425573122

Imbalanced speciation pulses sustain the radiation of mammals 2024
Quintero, I. , Lartillot, N., Morlon, H.
*Science **384** 1007-1012* 10.1126/science.adj2793

Phylogenetic Insights into Diversification 2024
Morlon, H., Andréoletti, J., Barido-Sottani, J., Lambert, S., Perez-Lamarque, B., **Quintero, I.**, Senderov, V., Veron, P.
*Annual Review of Ecology, Evolution, and Systematics **55** 1-21* 10.1146/annurev-ecolsys-102722-020508

The build-up of the present-day tropical diversity in tetrapods 2023
Quintero, I., Landis, M.J., Jetz, W., Morlon, H.
*Proceedings of the National Academy of Sciences **120** (20) e2220672120* 10.1073/pnas.2220672120

Population divergence associated with spatial asynchrony in precipitation in Neotropical frogs. 2022
Guarnizo, C.E., Montoya, P., **Quintero, I.**, Cadena, C.D.
Journal of Biogeography ***49*** *2169–2180* [10.1111/jbi.14484](10.1111/jbi.14484)

Macroevolutionary dynamics of climatic niche space. 2022
Quintero, I., Suchard, M.A., Jetz, W.
Proceedings of the Royal Society B ***289*** *20220091* 10.1098/rspb.2022.0091

Phylogenetic inference of where species spread or split across barriers. 2022
Landis, M.J., **Quintero, I.**, Muños, M.M., Zapata, F., Donoghue, M.J.
Proceedings of the National Academy of Sciences ***119*** *(13) e2116948119* 10.1073/pnas.2116948119

Nest architecture is linked with ecological success in songbirds. 2022
Medina, I., Perez, D., Silva, A., Cally, J., Leon, C., Maliet, O., **Quintero, I.**
Ecology Letters ***25*** *(6) 1365-1375* [10.1111/ele.13998](10.1111/ele.13998)

Global functional and phylogenetic structure of avian assemblages across elevation and latitude. 2021
Jarzyna, M., **Quintero, I.**, Jetz, W.
Ecology letters ***24*** *196-207* [10.1111/ele.13631](10.1111/ele.13631)

Interdependent trait and biogeographic evolution driven by biotic interactions. 2020
Quintero, I., Landis, M.J.
Systematic Biology ***69*** *(4) 739-755* 10.1093/sysbio/syz082

Global elevational diversity and diversification of birds. 2018
Quintero, I., Jetz, W.
Nature ***555*** *246-250* 10.1038/nature25794

Historical Biogeography Using Species Geographical Ranges. 2015
Quintero, I., Keil, P., Jetz, W., Crawford, F.W.
Systematic Biology ***64*** *(6) 1059-1073* 10.1093/sysbio/syv057

Asynchrony of seasons: genetic differentiation associated with geographic variation in climatic seasonality and reproductive phenology. 2014
Quintero, I., Gonzáles-Caro, S., Zalamea, P.C., Cadena, C.D.
The American Naturalist ***184*** *(3) 352-363* [10.1086/677261](10.1086/677261)

Rates of projected climate change dramatically exceed past rates of climatic niche evolution among vertebrate species. 2013
Quintero, I., Wiens. J.J.
Ecology Letters ***16*** *(8) 1095-1103* [10.1111/ele.12144](10.1111/ele.12144)

What determines the climatic niche width of species? The role of spatial and temporal climatic variation in three vertebrate clades. 2013
Quintero, I., Wiens. J.J.
Global Ecology and Biogeography ***22*** *(4) 422-432* [10.1111/geb.12001](10.1111/geb.12001)

Non-published Preprints

The Occurrence Birth-Death Diffusion Process: Unraveling Diversification Histories with Fossils and Heterogeneous Rates. 2025
Andréoletti, J., **Quintero, I.**, Morlon, H.
bioRxiv [10.1101/2025.08.26.672414](10.1101/2025.08.26.672414)

The Occurrence Birth-Death Diffusion Process: Unraveling Diversification Histories with Fossils and Heterogeneous Rates. 2024
Sanchez, A., **Quintero, I.**, Pedraza, S., Bonilla, D., Lohmann, L.G., Cadena, C.D., Zapata, F.
bioRxiv 10.1101/2024.05.28.596313

Software

Tapestree.jl *2022 – present*
Unraveling the evolutionary tapestry: Tapestree is a Julia package of phylogenetic analyses of diversification, trait and biogeographic dynamics. GitHub

Introduction to Julia *2019 – present*
A quick introduction to Julia focusing on efficviency. GitHub

rase *2015 – present*
Range Ancestral State Estimation for Phylogeography and Comparative Analyses. R package version 0.3-2. GitHub

Grants, Awards & Academic Distinctions

Tremplin-ERC StG from the Agence Nationale de Recherche (ANR) *2023*
project "The evolutionary tapestry: total-evidence interdependent models of species evolution" [€ 113,500].

The Australia and Pacific Science Foundation *2020*
project "Are open nests a key evolutionary innovation in birds?" with Iliana Medina [AUD 29,878].

Marie Sklodowska-Curie Action (MSCA-IF-2019) *2019*
project "INSANE Joint Species And Niche Evolution" [€ 184,707.84].

Yale's John Spangler Nicolas prize *2019*
for the best PhD Dissertation by an Ecology and Evolutionary Biology student [US$ 500].

Graduate Research Fellowship Program (GRFP) from the National Science Foundation (NSF) *2014*
PhD "The Evolutionary Tapestry: Insights on the Processes Behind the Origin and Maintenance of Biodiversity" [US$ 138,000].

Graduate Student Research Award from the Society of Systematic Biologists (SSB) *2014*
project "integrating species ranges into phylogeographic models" [US$ 2,000]

4th absolute place in Biological Sciences of 2010 nationally in Colombia's Graduate State Exam *2010*
Examen de Estado de calidad de la Educación Superior SABER PRO (antes ECAES), Colombia.

1st group nationally at Colombia's State Exam *2007*
Examen de Estado Para Ingreso a la Educación Superior, ICFES, Colombia.

Skills

Programming: Julia, R, Python, Git, bash, LaTex, Stan, INLA, GDAL

Languages: Spanish (native), English (fluent), French (C1)

Invited Talks & Academic Meetings

Invited lecturer Western Biology Evolution Seminar Series (WBESS) *Online Seminar series*
Imbalanced diversification sustain the rise, decline and fall of clades. *2026*

Invited lecturer at Spherical cows and bipedal goats: perspectives on mathematical models in biology — *École Normale Supérieure, Paris, France* — *2025*
Environmental controls on evolutionary dynamics using interdependent models.

Invited lecturer at the Chaire of the Mathematical Modelling and Biodiversity (MMB) — *Collége de France, Paris, France* — *2025*
Imbalanced diversification sustain the rise, decline and fall of clades.

Invited lecturer at the Seminar Series of the Centre de Recherches en Paléontologie (CR2P) — *Muséum National d'Histoire Naturelle, Paris, France* — *2025*
The rise, decline and fall of biodiversity.

Congress of the European Society for Evolutionary Biology (ESEB) — *Barcelona, Spain* — *2025*
1st talk: The rise, decline and fall of biodiversity. 2st talk: The diffused evolutionary dynamics of morphological novelty.

Invited lecturer by the PhDs and Postdocs at the Center for Macroecology, Evolution, and Climate (CMEC) — *University of Copenhagen, Denmark* — *2025*
The rise, decline and fall of biodiversity.

Invited lecturer at the BioGeoPhylo Workshop — *Aix-Marseille University, Luma Arles, Arles, France* — *2025*
On the effects of regional environment and modes of diversification.

Invited lecturer at the Seminar Series of the Department of Computational Biology — *University of Lausanne, Switzerland* — *2025*
The rise, decline and fall of biodiversity.

Invited lecturer at the Seminar Series Department of Biosciences — *Durham University, UK* — *2024*
The rise, decline and fall of clades.

Invited lecturer at L'Institut Diversité, Écologie et Évolution du Vivant (IDEEV) — *Université Paris-Saclay, Gif-sur-Yvette, France* — *2024*
Imbalanced speciation pulses sustain the radiation of mammals.

Speciation Gordon Research Seminar and Conference — *Renaissance Tuscany Il Ciocco in Lucca (Barga), Lucca, Italy* — *2023*
The birth-death diffusion leading to present-day Mammal diversity.

Invited lecturer at the Ecology Seminar Series of the Department of Ecology and Evolution — *(online) Texas A&M University-Corpus Christi, TX, USA* — *2022*
Insights into the processes behind the origin and maintenance of biodiversity.

MCEB: Mathematical and Computational Evolutionary Biology — *Porquerolles, France* — *2021*
Incorporating latent stochasticity into environmental, state dependent and general birth-death models.

Guest lecturer at Sebastian Höhna's lab seminar — *(online) Ludwig-Maximilians-Universität, Munich, Germany* — *2020*
Interdependent trait evolution with diversification and with biogeography.

AEEB : Evolutionary Biology Meeting at Marseille — *Marseille, France* — *2019*
Effect of fluctuating environment on spatial diversification.

Invited lecturer at the seminar series on Biodiversity — *Swiss Federal Research Institute, WLS, Birmensdorf, Switzerland* — *2019*
Insights on the processes behind the origin and maintenance of biodiversity.

International Society of Biogeography Biannual Meeting — *Malaga, Spain* — *2019*
Interdependent trait and biogeographic evolution driven by biotic interactions.

Evolution Annual Meeting — *Austin, TX, USA* — *2016*
Global altitudinal biodiversity of birds.

Ecological Society of America Annual Meeting — *Sacramento, CA, USA* — *2014*
Identifying the evolutionary and geographical underpinnings of global avian mountain biodiversity.

Phylotastic Hackathon — *Tucson, AZ, USA* — *2013*
www.phylotastic.org.

International Society of Biogeography Biannual Meeting — *Miami, FL, USA* — *2013*
Variation in avian diversification rate, species richness and elevation across mountain systems.

Smithsonian Tropical Research Institute BCI Talk — *BCI Island, Panamá* — *2012*
Asynchrony of Seasons: population differentiation resulting from spatial heterogeneity in climatic regimes?

III Symposium of Evolutionary Biology — *Medellín, Colombia* — *2011*
Asynchrony of Seasons: population differentiation resulting from spatial heterogeneity in climatic regimes?

Supervision

Master (M1) student Hugo Girault — *IBENS, Paris, France* — *Apr 2026 – July 2026*

Thesis director of doctoral researcher Paul Leroux — *IBENS, Paris, France* — *Jan 2026 – present*

PhD defense jury for Sami Touafchia — *Muséum National d'Histoire Naturel, Paris, France* — *Dec 2025*

Master (M1) student Niya Ren — *IBENS, Paris, France* — *Mar 2025 – June 2025*

PostDoc Dr. Olivia Price — *IBENS, Paris, France* — *Oct 2024 – Mar 2026*

Master (M2) student Thomas Merrien — *IBENS, Paris, France* — *Mar 2021 – July 2021*

Master (M2) student Clio Dalmaz — *IBENS, Paris, France* — *Nov 2019 – Feb 2020*

Undergraduate students Benjamin Kemplay & Adam Houston — *Yale University, CT, USA* — *2015*

Service as Reviewer

Nature, PNAS, Ecology Letters, Proceedings of the Royal Society B, Nature Ecology and Evolution, Nature Communications, Systematic Biology, Evolution, Journal of Animal Ecology, Evolutionary Ecology, PLoS One, NSF research grants. Publons

Workshops & Teaching

Introduction to probabilistic inference of Phylogenetic Comparative Methods (PCM) using Julia 2ed — *online* *2025*
6-days online workshop through Transmitting Science.

Introduction to Diversification models — *Université Paris Cité, Paris, France* *2025*
Part of Master's course with Dr. Didier Casane.

Diversification models — *Université Paris Cité, Paris, France* *2024*
part of module ACP, EvoGEM coordinated by Dr. Guillaume Achaz & Dr. Paul Zaharias.

Introduction to probabilistic inference of Phylogenetic Comparative Methods (PCM) using Julia 1ed — *online* *2023*
5-days online workshop through Transmitting Science.

Basic Introduction to Julia — *(online) Vancouver, Canada* *2022*
Workshop for the 10th Biennial conference of the International Biogeography Society.

Understanding macroevolutionary dynamics using RANDA and JPANDA — *online* *2022*
1-day online workshop through Transmitting Science with Dr. H. Morlon.

Introduction to Historical Biogeography — *online* *2021*
part of Master course "Biology of Ecological Systems (Biologie des Systèmes Écologiques)".

Julia Practical — *École Normale Supérieure, Paris, France* *2021*
part of Master course "Computational Biology"

Introduction to Historical Biogeography — *(online) École Normale Supérieure, Paris, France* *2020*
part of Master course "Biology of Ecological Systems (Biologie des Systèmes Écologiques)".

Introduction to Historical Biogeography — *(online) Laboratoire d'Ecologie Alpine (LECA), Grenoble, France.* *2020*
Gradiate level course "EcoSum: Ecological Models and Beyond".

Introduction to Julia — *Yale University, CT, USA* *2017*
Yale Center for Research Computing Bootcamp.

Teaching Fellow for Evolutionary Biology — *Yale University, CT, USA* *2015*
Professor: Dr. J. Powell and Dr. S. Stearns.

Guest Lecturer for Introductoery Statistics: Life Sciences — *Yale University, CT, USA* *2014*
Introduction to Likelihood and Bayesian Inference. Professor: Dr. W. Jetz.

Teaching Fellow for Introductory Statistics: Life Sciences — *Yale University, CT, USA* *2014*
Professor: Dr. W. Jetz.

Teaching Fellow for General Ecology *Yale University, CT, USA*
Professors*:* Dr. D. Post & Dr. D. Vasseur. *2014*

Teaching Assistant for Vertebrates Theory Class *Universidad de los Andes, Bogotá, Colombia*
Professors*:* Dr. C.D. Cadena. *2011*

Teaching Assistant for Animal Physioogy Laboratory *Universidad de los Andes, Bogotá, Colombia*
Professors*:* Dr. A. Amézquita. *2010*

Reference Contacts

- Hélène Morlon. CNRS research director, IBENS, École Normale Supérieure, UMR 8197 CNRS. 46 rue d'Ulm, 75005 Paris, France. e-mail. +33 (0)1 44 32 35 35.
- Michael Landis. Associate Professor, Department of Biology, Washington University in St. Louis. One Brookings Dr., 63130 St. Louis, MO, USA. e-mail. +1 (314) 935 8082.
- Walter Jetz. Associate Professor, Department of Ecology and Evolutionary Biology, Yale University. 165 Prospect St., 06511 New Haven, CT, USA. e-mail. +1 (203) 432 7540.
- Carlos Daniel Cadena. Professor, Departamento de Ciencias Biológicas, Universidad de los Andes. Cra. 1 No. 18a-12, 111711 Bogotá Colombia e-mail. +57 (1) 339 4949 ext. 2072.